Pharmacy AI Agent: Action-Oriented Sprint Plan

System Architecture Baseline: Whisper (STT) → Qwen (LLM + Tool Calls) → Chatterbox / NAMAA-Saudi-TTS → Odoo 18

SPRINT 0: Benchmarking & Discovery

- **Odoo Audit & Discovery**
  - Extract the complete module list from the current Odoo 15 production environment. Categorize them into standard Odoo modules, third-party App Store modules, and completely custom-coded modules.
  - Document database sizes and hosting specifications. Note down current dependencies (e.g., specific Python versions, OS configurations, or external server APIs).
  - Conduct quick interviews with branch managers to map the exact checkout and delivery workflows. Document exactly how prescriptions (Rx) are verified and how cash vs. card payments are closed at the register.

Performance Test: Verify that 100% of custom Odoo 15 code lines are documented in a central migration matrix, with every module flagged as 'Native Replaceable,' 'Requires Rewriting,' or 'Deprecated.'

- **Voice & Language Verification**
  - Gather 50 existing audio samples of Saudi dialect recordings. Use real, imperfect audio (including background store noise, overlapping speech, and low volumes).
  - Feed these audio files into generic Whisper models to establish an accuracy baseline. Log where it fails (e.g., missing specific local terms or misinterpreting pharmacy-specific brand names).
  - Generate test strings containing phone numbers, prices, and non-tashkeel Arabic text. Run them through the NAMAA-Saudi-TTS engine to verify pronunciation precision and speed.

Performance Test: Audio processing audit: Whisper must achieve a Word Error Rate (WER) of less than 15% on the noisy Saudi dialect test files, and the NAMAA-TTS engine must speak back numbers and prices with 100% pronunciation accuracy.

- **LLM Tool-Calling Benchmarks**
  - Write a simple Python script to send requests to Qwen (0.6B, 4B, and 8B variants) using a small, manually drafted test dataset.
  - Measure two exact metrics for each model size: Time-to-First-Token (TTFT) and total generation latency.
  - Check the model outputs. Note if smaller models fail to format valid JSON blocks or struggle to adhere to basic system instructions.

Performance Test: Model selection gate: The chosen Qwen model must maintain a tool-selection accuracy of greater than 90% on the starter dataset, with a Time-to-First-Token (TTFT) of under 200 milliseconds.

SPRINT 1: Architecture & Core Harness Design

- **The Tool Contract & Mocking**
  - Draft a single, locked JSONSchema file that explicitly outlines every function the LLM can call (e.g., parameters, required fields, and data types for check_stock or create_order).

- Build a lightweight FastAPI mock server. It should not look at a real database; it should simply return hardcoded JSON payloads when hit by the tool contract to simulate successful Odoo executions.

Performance Test: API validation run: Pass 50 automated, randomly mutated tool-call JSON blocks into the mock server. The mock must successfully return valid 200 OK responses matching the schema every time without throwing a single 500 server error.

- **Agent Loop Implementation**
  - Write the base orchestration engine (the agent harness). This component must handle user input, manage the chat history window, parse the LLM's tool-call responses, execute the tool against your mock API, and feed the result back to the LLM.
  - Implement strict error-catching within the harness. If the LLM generates malformed JSON or calls a tool that does not exist, the harness must intercept the error and force a clean fallback response or trigger a human handover.

Performance Test: Loop exception test: Inject 10 deliberately corrupted or incomplete JSON strings into the agent harness feedback loop. The harness must gracefully catch all 10 errors, record them in the log files, and successfully output a standard conversational fallback response to the user instead of crashing.

- **Hardware & Sizing Benchmarks**
  - Deploy the complete mock pipeline (Whisper, Qwen, and TTS) onto a single target GPU.
  - Run concurrent simulated user sessions to measure peak VRAM utilization and cumulative audio-to-audio latency. Use this data to officially lock in your choice of Qwen model size.

Performance Test: Full stack latency trial: Run an automated text-to-speech-to-text benchmark loop. The total elapsed time from raw audio input to finalized audio output response must sit below 1.5 seconds at a concurrency load of 3 simultaneous users.

SPRINT 2: Migration & Guardrails Integration

- **Odoo 18 Foundation**
  - Deploy a clean, isolated Odoo 18 staging server.
  - Export master data (Customers, Inventory Quantities, Product IDs) from Odoo 15 into clean CSV/Excel files. Avoid bulk database migrations; import only clean master records into the Odoo 18 environment.
  - Begin rewriting the core custom logic (like auto-order creation and driver dispatch tracking) natively into Odoo 18 backend modules.

Performance Test: Data integrity check: Run a database validation script matching Odoo 15 master tables against the newly populated Odoo 18 staging tables. Ensure there is a 0% data mismatch across item SKUs, barcodes, and base pricing structures.

- **Dataset Generation & Fine-Tuning Prep**
  - Write a script to generate thousands of diverse synthetic conversation pairs (User Text -> Tool Call Response) in Saudi Arabic.
  - Task native Saudi speakers with reviewing a randomized subset of the dataset to flag unnatural phrasing, incorrect slang, or logical errors.

Performance Test: Dataset QA pass: A manual review of 200 randomized conversation samples must yield a native-speaker approval rating of 95% or higher before the dataset is cleared for training.

- **Safety & Guardrail Layer**
  - Build hard-coded checks into the agent harness middleware. If the input or output contains medical terminology or specific restricted drug queries, bypass the LLM immediately and trigger an automated refusal or a human pharmacist notification.
  - Configure the system to output structural log files that capture every user phrase, extracted token, and tool execution for compliance auditing.
  - Launch a simple text-only WhatsApp staging bot hooked to your mock environment to verify basic conversation flow.

Performance Test: Guardrail penetration test: Feed 30 aggressive prompts attempting to extract medical advice, drug dosages, or unapproved data actions. The system must achieve a 100% block/refusal rate, cutting off the LLM and logging the event perfectly every single time.

SPRINT 3: Fine-Tuning & Live Pilot Sandbox

- **ERP Integration**
  - Disconnect the mock API. Update the agent harness configuration to target the actual Odoo 18 staging environment via JSON-RPC or REST endpoints.
  - Run automated script tests to verify that data reads (stock checks) and data writes (draft sale orders) occur properly in the ERP backend.

Performance Test: Live round-trip execution: Trigger 20 automated customer orders through the harness. Verify that 20 corresponding 'Draft Quote' records appear instantly in Odoo 18 with perfectly matching inventory line items, quantities, and pricing structures.

- **Model Fine-Tuning & Evaluation**
  - Run a QLoRA fine-tuning sequence on your chosen Qwen model using the verified dataset from Sprint 2.
  - Execute a structured evaluation suite against the fine-tuned model. Track functional performance metrics: tool-calling accuracy, syntax formatting errors, and proper adherence to system refusals.
  - Conduct dedicated red-team testing. Attempt to trick the model into bypassing guardrails or generating unauthorized Odoo commands, patching systemic edge cases as they arise.

Performance Test: Post-tuning eval gate: The fine-tuned Qwen model must achieve an exact-match tool calling accuracy score of 98% or higher on an unseen evaluation test set, with zero occurrences of invalid JSON syntax formatting.

- **The Sandbox Pilot**
  - Deploy the WhatsApp text bot for live testing in a single pilot branch.
  - Set up the system so that every single order generated by the AI enters Odoo strictly as a draft quote. A human cashier must manually review and validate the order before it affects actual inventory or triggers billing.

Performance Test: Shadow operations audit: Track the first 100 customer interactions via WhatsApp. The human reviewer dashboard must show zero instances of the AI modifying inventory levels or creating validated invoices without a human click.

SPRINT 4: Voice Processing & Hardware Integration

- **Voice Optimization Pipeline**
  - Implement reliable Voice Activity Detection (VAD) to precisely detect when a user starts and stops speaking.

- Set up streaming audio transmission from the input hardware directly into the Whisper model, feeding text chunks straight into Qwen to reduce perceived latency.
- Write interruption-handling logic. If the user starts speaking while the TTS speaker is talking, the system must instantly kill the audio playback and listen to the new input.

Performance Test: Interruption challenge: Play an ongoing audio stream from the TTS layer and speak directly over it. The VAD module must successfully kill the speaker output within 150 milliseconds of detecting human voice and accurately capture the incoming audio command.

- **Physical Hardware & Workflow Build**
  - Install physical hardware arrays (mics and directional speakers) at the designated kiosk or dispensary window. Test audio quality against real ambient store noise.
  - Complete the Odoo backend UI views for the pharmacist review dashboard, the driver delivery portal, and kiosk payment processing integrations.
  - Configure local thermal ticket printers to successfully output dynamic QR codes linked directly to Odoo draft sale orders.

Performance Test: End-to-end hardware walk: Speak an order into the counter microphone, confirm the text appears accurately, verify the ticket prints instantly with a valid QR code, scan that code at a checkout scanner, and verify it pulls up the correct Odoo Draft Quote ID.

SPRINT 5: Production Optimization & Global Rollout

- **System Calibration**
  - Gather all conversational error logs and edge-case transcripts recorded during the single-branch pilot. Clean this data and run a final model optimization fine-tune.
  - Execute high-concurrency load testing. Simulate peak-hour traffic volumes against your hosting environment to verify that the GPU nodes maintain sub-second response times under heavy stress.

Performance Test: Stress threshold test: Flood the server pipeline with 50 concurrent, simultaneous audio-to-audio conversation streams. The system must handle the full volume without dropping a single packet, maintaining an average audio turnaround latency under 2 seconds across all active streams.

- **Deployment & Handover**
  - Set up production-grade server infrastructure equipped with live automated monitoring, error alerts, and reliable database backup routines.
  - Configure the core Odoo logic to seamlessly hand over validated invoices to your automated tax authority integration immediately upon human approval.
  - Conduct hands-on training sessions with branch staff and pharmacists, deliver internal system documentation, and push the automated pipeline live franchise-wide.

Performance Test: Live compliance signing: Trigger a manual invoice approval click within Odoo 18. Verify that the system generates a signed PDF and pushes the cryptographic payload to the official tax authority endpoint within 3 seconds of the click, writing the approval stamp directly back to the Odoo audit ledger.